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CSE 412
Assignment 4

What the project does

I built a small Flask web app that runs the same search query two ways: once with an index and once without. The dataset is a fake stock trade ledger I generated with Python. The point is to see how much faster Postgres gets when it can use a btree index instead of doing a full table scan.

Results

The screenshots below show the UI side by side with the same searches. Same rows are returned in both cases (which is what you'd expect - the index doesn't change the result, it just changes how Postgres finds it).

- The single-table query benefits a lot because there are 280 different tickers in the data, so AAPL only matches a small fraction of rows. The bitmap index scan jumps straight to those rows. Without the index it has to look at every row in the trades table.
- The join benefits even more because the broker filter on traders is also selective (200 brokers), and the planner can use a nested loop with the trader_id index instead of a hash join over a full sequential scan.
- Re-running the same query a few times made the times more stable because of Postgres's shared buffer cache. The numbers above were taken after the cache was warm.

How to run it

Read the README.

Screenshots

CSE 412 Assignment 4 - Indexed vs Non-Indexed Queries

Stock trade ledger. Try ticker AAPL, GOOGL, TSLA, etc. Try broker Fidelity, Schwab, Robinhood, etc.

Search without indexing

Single table (trades by ticker)

AAPL

Search

Execution time: 2.382 ms

trade_id	trader_id	ticker	side	quantity	price	executed_at
1611	16783	AAPL	BUY	74	107.4299	2026-04-04 17:24:15.618274
3425	22535	AAPL	BUY	390	262.0304	2026-04-01 02:20:15.618274
4894	22249	AAPL	BUY	59	14.3899	2026-03-31 12:49:12.618274
1184	16944	AAPL	BUY	85	657.5270	2026-03-19 21:19:45.618274
730	12561	AAPL	SELL	460	143.2802	2026-03-13 04:52:04.618274

Join (trades + traders by broker)

Fidelity

Search

Execution time: 11.319 ms

trade_id	full_name	broker	ticker	side	quantity	price	executed_at
4145	Elizabeth Martin	Fidelity	AWX	BUY	247	699.9185	2026-04-06 08:11:42.618274
1597	Jennifer Martin	Fidelity	DVX	SELL	831	102.9602	2026-04-05 22:34:11.618274
2911	Charles Martinez	Fidelity	GS	BUY	463	425.8067	2026-02-21 10:14:40.618274
1022	Lukas Davis	Fidelity	EHZ	SELL	416	152.8796	2026-01-21 09:02:15.618274
4944	Robert Rodriguez	Fidelity	ELX	BUY	300	660.0985	2025-11-09 01:03:06.618274

Search with indexing

Single table (trades by ticker)

AAPL

Search

Execution time: 0.529 ms

trade_id	trader_id	ticker	side	quantity	price	executed_at
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3425	22535	AAPL	BUY	390	262.0304	2026-04-01 02:20:15.618274
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Search without indexing

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Single table (trades by ticker)

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Search With Indexing

Search with Indexing

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